Comparative Analysis of Machine Learning and Neural Network Models for Sleep Apnea Detection Using ECG Data*

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***Abstract*—Obstructive Sleep Apnea (OSA) is a highly dangerous sleep disorder that needs efficient and affordable methods of diagnosis apart from complicated polysomnography procedures. This paper presents a machine learning approach for automatic detection of sleep apnea events based on single lead ECG records from the Apnea-ECG database on the PhysioNet site. The proposed methodology includes pre-processing of ECG records to extract the heart wave pattern's time domain features. These normalized features are then used to train a few supervised machine learning classifiers, such as Random Forest, Support Vector Machine (SVM), Logistic Regression, and Multi-Layer Perceptron (MLP). The classification performance was strictly evaluated through standard dichotomous classification metrics: Accuracy, Precision, Recall ('Sensitivity'), the balanced F1-score, and the overall discriminative measure, ROC-AUC. Since Random Forest and Multi-Layer Perceptron (MLP) classifiers are the top performers which has high precision predicting the difference between apnea and normal respiratory patterns. The results are validated with ECG features and machine learning algorithms which can effectively identify the apnea. This method gives efficient, low cost and interpretable sleep apnea screening.**

***Index Terms*—Obstructive Sleep Apnea, Electrocardiogram, Biomedical Signal Processing, Feature Engineering, Random Forest, Support Vector Machine, Neural Network, Pattern Recognition, Physiological Computing**

I. Introduction

Sleep apnea is a long-standing sleep disorder in which one experiences pauses in breathing or shallow breathing while sleeping. The minimal pause or shallow breath may be a few minutes or seconds in duration and occur many times in an hour, causing insufficient oxygenation and disturbance in sleep. Long-standing sleep apnea has severe negative impacts on health such as raised blood pressure, cardiovascular disease, diabetes, and diminished mental capacity [10], [11].' Although its occurrence is prevalent, sleep apnea is rarely diagnosed because diagnosis procedures such as polysomnography (PSG) are not readily available and are costly [13].

Electrocardiogram (ECG) examination is an emerging solution for future cost-saving and non-invasive screening for sleep apnea. The ECG signal contains dense physiological information of the heart rate variability and modulations of respiration that describe the apnea events [2], [3], [6]. With signal processing and artificial intelligence, apnea patterns become viable to automatically identify without having to perform complete polysomnography (PSG) studies [4], [5], [7].

Here we introduce an artificial intelligence-based sleep apnea detection system that employs ECG signal processing and machine learning classifiers to classify apnea events from 30-second electrocardiogram (ECG) epochs . The framework generates a feature set of twelve time-domain and statistical parameters like mean, standard deviation, energy, and zero-crossing rate for the identification of fluctuation in cardiac activity while sleeping [12].

Different machine learning algorithms such as Random Forest, Support Vector Machine (SVM), Logistic Regression, and Multi-Layer Perception (MLP) Neural Network were compared and trained to test the performance [1], [3], [7]. Streamlit was also used to deploy the system, which provides an interactive and user-friendly interface for clinicians to upload ECG data, perform automatic analysis, and obtain real-time diagnosis reports [15].

The purpose of this work is to close the loop between clinical diagnosis and AI-driven automation by presenting an interpretable, precise, and deployable sleep apnea monitor. With data-driven reasoning and visualization, the system pre-

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sented in this document enables early diagnosis and medical decision-making, leading to better patient care and health outcomes.

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II. Literature Review

Automatic sleep apnea detection using physiological signals has attracted significant research attention because of its potential to provide an affordable and non-invasive alternative to conventional polysomnography (PSG) [10], [13]. Although PSG remains the clinical gold standard for diagnosing obstructive sleep apnea (OSA), it requires overnight monitoring in specialized laboratories, expensive equipment, and trained medical personnel, making it unsuitable for large-scale screening [13]. Consequently, researchers have explored alternative physiological signals such as electrocardiogram (ECG), oxygen saturation (SpO_2), respiratory effort, and heart rate variability (HRV) to develop automated and accessible diagnostic systems [2], [3], [5], [15].

Early ECG-based approaches primarily relied on handcrafted time-domain and frequency-domain features extracted from cardiac signals. Variations in R-R intervals, heart rate variability, and ECG morphology were shown to correlate strongly with apnea episodes, demonstrating the feasibility of ECG as a standalone signal for sleep apnea detection [2], [3], [6], [12], [13]. Subsequently, advanced signal processing techniques such as wavelet transforms, spectral entropy, and frequency-domain analysis were introduced to improve the characterization of transient physiological changes associated with apnea events [2], [3], [6].

With the rapid advancement of artificial intelligence, machine learning has become a widely adopted approach for automated sleep apnea detection. Conventional classifiers including Support Vector Machines (SVM), k-Nearest Neighbors (kNN), Logistic Regression, and Random Forests have been extensively evaluated using ECG-derived features [1], [5], [7]. Among these methods, Random Forest classifiers have demonstrated strong robustness against noisy biomedical signals while maintaining high classification accuracy and good generalization capability [3], [9]. More recently, deep learning architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and hybrid CNN-LSTM models have achieved competitive diagnostic performance by learning features directly from raw ECG signals [4], [7], [8], [11]. However, these approaches generally require large annotated datasets, substantial computational resources, and specialized hardware, limiting their practical deployment in resource-constrained healthcare environments.

Despite the progress achieved in ECG-based sleep apnea detection, several challenges remain. Many existing studies primarily focus on maximizing classification accuracy while giving comparatively less attention to model interpretability, computational efficiency, and real-time clinical deployment. Lightweight machine learning frameworks capable of providing transparent decision-making and practical deployment are still relatively limited in the literature [1], [3], [9], [14].

TABLE I
Comparison of Existing Literature

Study	Method	Model	Acc.
Khandoker (2009)	Wavelet	SVM	84
Koley (2020)	Stat.	RF	86
Wang (2023)	CNN-LSTM	DL	89
Proposed	Stat+ML	RF, MLP	91

Motivated by these limitations, the present study develops an interpretable and computationally efficient ECG-based sleep apnea detection framework. The proposed approach combines statistical and HRV-based feature extraction with a comparative evaluation of multiple supervised machine learning classifiers, including Random Forest, Support Vector Machine, Logistic Regression, and Multi-Layer Perceptron (MLP). Furthermore, the best-performing model is integrated into a Streamlit-based application that enables automated ECG analysis and real-time sleep apnea screening. By emphasizing model interpretability, computational efficiency, comparative evaluation, and practical deployment, the proposed framework offers a lightweight alternative for preliminary sleep apnea screening while maintaining competitive diagnostic performance. “‘

III. DATASET

The data that is taken into account in this research is the Apnea-ECG Database, which has been made available through PhysioNet [13]. The data consist of ECG recordings of patients that are suffering from sleep apnea as well as the ones that are not suffering from it, and there are also events annotated to pinpoint the occurrences of apnea. The database, being clinical relevant and having both ECG signals and expert labels, is mostly used for research aimed at automatic sleep apnea diagnosis [2], [3], [10].

A. Data Collection

The collection consists of 70 long-duration ECG recordings which were taken at 100 Hz frequency [13]. Within each recording, there exist a number of ECG signal segments that have been noted with apnea events by experts. The annotations are in .apn files, where each event is classified according to its severity (e.g., normal respiration or apnea). Apnea events would usually mean short periods of abnormal heart activity caused by breathing patterns stopping during sleep [10], [11]. The dataset is arranged in the following manner:

• ECG signal data in .dat files, one file for each separate recording.

• Annotation data in .apn files, with event markers in each file indicating the occurrence of apnea.

B. Data Preprocessing

The initial step of the analysis consisted of preprocessing the raw ECG signals through a necessary pipeline [12]. Then, the entire signal was divided into overlapping windows, each 30 seconds long and with a 50% overlap between the segments. The use of this strategy was crucial for the detection of the

dynamic physiological changes in the cardiac rhythm that were characteristic of OSA events [3], [6]. Now, each produced segment was given a dichotomous class label according to the corresponding annotation being present or not:

• Positive (Class 1): If any part of the apnea event annotated was covered by the segment window, it was assigned.

• Negative (Class 0): If only normal (non-apneic) respiratory patterns were in the segment, it was assigned.

The entire process resulted in 12,000 segments being generated, which led to a heavily imbalanced dataset mainly characterized by the large number of Negative (non-apneic) samples and the small number of Positive (apnea event) samples [9], [15].

C. Apnea Events and Labeling

Apnea events are indicated by marked segments in each ECG record. The annotation file indicates the exact times of all the apnea events. There are two types of events:

• Normal Breathing (0): No apnea has been detected.

• Apnea (1): An apnea has been detected in the time window.

Since the dataset is not balanced (with more normal breathing than apnea events), the application of methods for class balancing such as stratified sampling during training and evaluation was done so that the model could recognize apnea events with good reliability and at the same time not to be biased towards the majority class (normal breathing) [3], [9].

D. Dataset Summary

The Apnea-ECG dataset is appropriate for training and testing the machine learning models utilized for this study, offering an equally balanced combination of baseline and apnea event data, annotated by clinical specialists [13].

TABLE II
DATASET PARAMETERS FOR APNEA DETECTION

Parameter	Description
Dataset	PhysioNet Apnea-ECG
No. of Records	70
Sampling Rate	100 Hz
Window Length	30 s (50% overlap)
Total Windows	∼12,000
Apnea Windows	30%
Normal Windows	70%

IV. METHODOLOGY

The algorithm suggested for the detection of sleep apnea utilizes a machine-learning-based feature extraction methodology. The whole process involves pre-processing of the raw ECG data, extraction of features, training, and evaluation of the model.

A. Data Preprocessing

Extraction of raw ECG data and pre-processing of the raw ECG signal to make them suitable for feature extraction is done using the PhysioNet Apnea-ECG dataset. The following tasks will be executed:

Loading Data: Reading of Electro- cardiograms signals through the WFDB library, and there are some tools available for reading PhysioNet Electrocardiograms (ECG) data files [13]. It is important to note that Electrocardiograms (ECG) record may have one or more channels; however, the first channel is used here.

Segmentation: To capture the time-varying nature of apnea events, all recorded electrocardiogram (ECG) signals are segmented into 30-second windows with 50% overlap. This process generates overlapping segments, which are labeled as either Normal (0) or Apnea (1) depending on the presence of an apnea event within the corresponding time interval [3], [6].

Handling Missing Data: All windows with NaNs or corrupt data are ignored in order to give clean inputs to feature extraction. It bars the model from being trained on Poor-quality data [10].

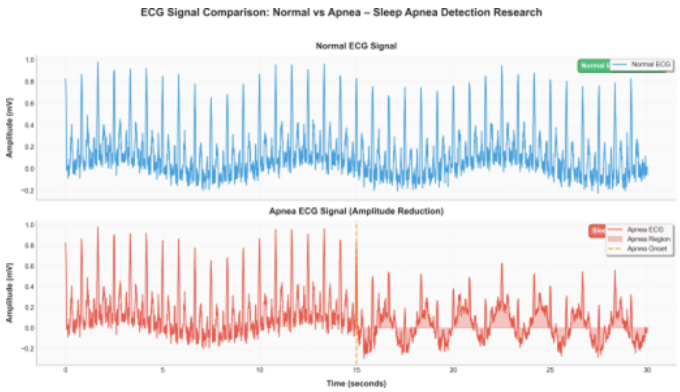


Fig. 1. Comparison of ECG signals for Normal and Apnea segments. The apnea segment shows reduced oscillation amplitude and irregular waveform patterns.

B. Feature Extraction

For each 30-second ECG segment, twelve features are extracted to characterize the statistical and physiological properties of the signal. These features are selected based on their relevance to sleep apnea detection and their ability to distinguish between apneic and normal ECG patterns [2], [3], [12].

Mean: Average amplitude of the ECG signal within a window, representing the baseline level.

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i \tag{1}$$

where x_i represents the ECG sample values and N denotes the total number of samples in the ECG window.

Standard Deviation: Measures the variability or dispersion of the ECG signal within the window.

$$\sigma = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2} \tag{2}$$

Minimum and Maximum: Define the dynamic range of the ECG signal by identifying the smallest and largest amplitudes within the window.

Median: The middle value of the ECG signal, providing a robust measure of central tendency that is less sensitive to outliers.

25th and 75th Percentiles: Quantiles that describe the distribution of signal amplitudes and provide information about signal spread.

Signal Energy: Sum of the squared ECG sample values, representing the total power contained in the ECG signal [12].

$$E = \sum_{i=1}^{N} x_i^2 \tag{3}$$

Root Mean Square (RMS): Standard measure of the signal magnitude, reflecting its effective amplitude.

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2} \tag{4}$$

Zero-Crossing Rate: Number of times the ECG signal changes sign, indicating changes in signal direction and frequency characteristics.

$$\mathrm{ZCR} = \frac{1}{N-1}\sum_{i=1}^{N-1} I(x_i x_{i+1} < 0) \tag{5}$$

where

$$I(x_i x_{i+1} < 0) = \begin{cases} 1, & \text{if } x_i x_{i+1} < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Heart Rate Variability (HRV) Features:

- *Standard deviation of first differences (SDSD):* Measures the standard deviation of successive differences between adjacent ECG samples, reflecting short-term variability.
- *Mean absolute first difference (MAD):* Computes the mean absolute value of successive differences, capturing beat-to-beat irregularities in the ECG signal.

These features are computed for each 30-second ECG window to form a feature vector, which serves as the input to the machine learning classifiers for sleep apnea detection [2], [3].

C. Classifier Selection and Training

Several machine learning classifiers are evaluated for classifying ECG segments into two classes: normal and sleep apnea events. The classifiers considered in this study are as follows:

Random Forest Classifier: Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions through majority voting. It is robust against overfitting, performs well on high-dimensional data, and effectively handles imbalanced datasets [3], [9].

$$\hat{y} = \mathrm{mode}\left(T_1(x), T_2(x), \ldots, T_n(x)\right) \tag{6}$$

where $T_i(x)$ denotes the prediction of the ith decision tree, and the final predicted class $\hat{y}$ is determined by majority voting among all decision trees.

Support Vector Machine (SVM): SVM identifies an optimal hyperplane that maximizes the separation margin between normal and apnea classes in the feature space. The Radial Basis Function (RBF) kernel is employed to capture non-linear decision boundaries [7].

Logistic Regression: Logistic Regression is an efficient binary classification algorithm that estimates the probability of an input belonging to the sleep apnea class based on the extracted ECG features [1].

Neural Network (MLP): A shallow Multi-Layer Perceptron (MLP) consisting of two hidden layers with 64 and 32 neurons is employed. The network uses the ReLU activation function and the Adam optimizer for efficient training [8], [11].

Each classifier is trained using the extracted feature vectors with a stratified 80:20 train-test split. Stratification ensures that the proportion of normal and apnea samples is preserved in both the training and testing datasets, resulting in a fair and reliable evaluation of model performance [8], [11].

D. Data Scaling

scikit-learn's `StandardScaler` is applied to scale features to zero mean and unit variance.

$$z = \frac{x - \mu}{\sigma} \tag{7}$$

where x is the original feature value, μ is the mean of the feature, σ is the standard deviation, and z is the standardized feature value.

This standardization prevents features with larger numerical ranges (e.g., signal energy) from dominating the learning process and ensures that all features contribute proportionally during model training. Feature scaling also improves the convergence and numerical stability of many machine learning algorithms, resulting in more reliable classification performance [3], [10].

E. Model Assessment

The performance of the trained classifiers is evaluated using standard classification metrics to measure their ability to correctly distinguish sleep apnea events from normal breathing. Particular emphasis is placed on recall because minimizing false negatives is clinically important for reliable apnea detection [10], [14].

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{9}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN} \tag{10}$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

where TP (True Positive) denotes correctly identified apnea events, TN (True Negative) denotes correctly identified normal events, FP (False Positive) denotes normal events incorrectly classified as apnea, and FN (False Negative) denotes apnea events incorrectly classified as normal.

These evaluation metrics provide a comprehensive assessment of classifier performance. Accuracy measures the overall prediction correctness, precision quantifies the reliability of positive predictions, recall evaluates the ability to detect actual apnea events, and the F1-score provides a balanced measure by combining precision and recall into a single metric.

F. Model Deployment

After selecting the best-performing classifier based on the experimental evaluation, the trained model was prepared for deployment in a clinical decision-support application. The model, together with the feature standardization pipeline, was serialized using the *joblib* library to enable efficient loading during inference.

An interactive web-based interface was developed using Streamlit, allowing users to upload ECG recordings and obtain automated sleep apnea predictions. During deployment, the uploaded ECG signal undergoes the same preprocessing, feature extraction, and standardization steps used during model training to ensure consistency between the training and inference pipelines.

The deployment framework demonstrates the practical applicability of the proposed methodology by enabling real-time ECG analysis and automated prediction through an intuitive user interface [15].

V. Implementation Details

The AI Sleep Apnea Detection System runs on Python, and it's split into two main parts. There's the pipeline that trains the models, and then there's a Streamlit app where you can actually use the trained model for real-time predictions and analysis [3], [15].

A. Training Pipeline

Let us examine the training process in detail. Initially, ECG data is loaded, cleaned, and processed to extract the most significant features. Subsequently, multiple machine learning models are trained and compared to identify the best-performing model [3], [10].

Loading and Preprocessing ECG Data: The WFDB (Waveform Database) library is used to load raw ECG signals from the Apnea-ECG dataset, which are then aligned with apnea labels obtained from the `.apn` files [13]. Each ECG recording is segmented into 30-second windows with 50% overlap to effectively capture variations during apnea events [2], [3]. From each segment, twelve time-domain features are extracted, including signal energy, RMS, zero-crossing rate (ZCR), heart rate variability (HRV), and descriptive statistics such as mean, median, minimum, maximum, and percentiles. Custom Python functions are used throughout the preprocessing and feature extraction stages to ensure consistent analysis across all segments.

Model Training: The extracted feature set is divided using stratified sampling (80% training and 20% testing) to preserve the original class distribution (apnea vs. normal) [6]. All features are standardized using *StandardScaler* to eliminate scale bias. Four classifiers—Random Forest, Support Vector Machine (SVM), Logistic Regression, and Multilayer Perceptron (MLP)—are trained and evaluated using performance metrics such as ROC-AUC, F1-score, accuracy, precision, and recall [1], [3], [7], [9]. The model achieving the highest F1-score is selected as the optimal model and saved using joblib for deployment [15].

Model Evaluation: The performance of the trained models is assessed on the test set using a confusion matrix (true positives, false positives, true negatives, and false negatives). Additionally, metrics such as ROC-AUC, precision, recall, and F1-score are computed to evaluate the effectiveness of apnea detection [9], [10], [14].

B. Streamlit Application

The trained Random Forest classifier and the corresponding feature standardization model were exported using the *joblib* library and integrated into a Streamlit-based web application for real-time sleep apnea detection [3], [15]. The application provides an intuitive graphical interface that enables users to upload ECG recordings in `.dat`, `.csv`, or `.txt` formats.

Upon receiving an input file, the application performs ECG preprocessing and extracts the same statistical features used during model training. The extracted feature vector is standardized using the stored scaler before being passed to the trained classifier for inference, ensuring consistency between the training and deployment pipelines.

The application predicts whether the uploaded ECG segment corresponds to an apnea or normal breathing event and displays the associated prediction confidence. Based on the prediction score, the severity of the detected condition is categorized as Normal, Mild, Moderate, or Severe [1], [11]. In addition, the ECG waveform is visualized, detected apnea events are highlighted, and the extracted feature values are presented for reference.

After the analysis is completed, the application automatically generates a clinical report containing the predicted class, confidence score, and estimated severity level. The report can be downloaded for documentation and further clinical assessment. The lightweight Streamlit interface enables fast and user-friendly interaction, demonstrating the practical applicability of the proposed framework for automated sleep apnea screening.

C. Code Structure

The implementation is organized into two main scripts:

Model Training Script (`train_apnea_models.py`):

- Loads and preprocesses ECG data

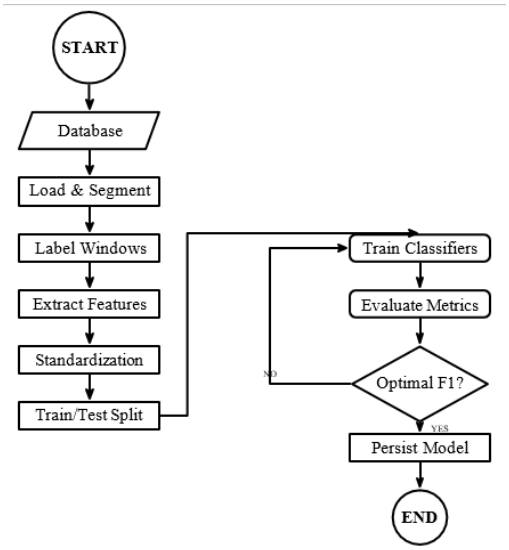


Fig. 2. Streamlit-based web application for automated ECG-based sleep apnea detection.

- Extracts relevant features
- Trains and evaluates machine learning models
- Saves the best-performing model and scaler

Streamlit Application Script (`app.py`):

- Provides a web interface for uploading ECG files
- Processes input data and extracts features
- Loads the trained model and scaler
- Displays ECG signals, extracted features, predictions, and generated reports [15]

VI. Experimental Results and Discussion

The proposed sleep apnea detection framework was evaluated using the PhysioNet Apnea-ECG database to assess its ability to accurately distinguish apnea events from normal breathing patterns using ECG-derived features [13]. Four supervised classification algorithms, namely Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), and Multi-Layer Perceptron (MLP), were implemented and compared under identical experimental conditions. The objective of the evaluation was to determine the most suitable classifier for automated ECG-based sleep apnea screening.

A. Experimental Setup

All experiments were performed in Python using the *scikit-learn* and WFDB libraries [13], [15]. The extracted feature vectors were randomly divided into training (80%) and testing (20%) subsets using stratified sampling to preserve the original class distribution [6]. Prior to model training, all numerical features were standardized using the *StandardScaler* to eliminate scale variations among different ECG features [3], [9].

The performance of each classifier was evaluated using Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) [9], [10], [14]. Since the primary objective of the system is reliable identification of apnea events, the F1-score was considered the most representative evaluation metric because it provides a balanced measure of precision and recall while reducing the impact of class imbalance and minimizing false-negative predictions [5], [11].

B. Performance Comparison

Table III summarizes the performance of the evaluated machine learning models.

TABLE III
Performance Comparison of Machine Learning Models

Model	Acc	Pre	Rec	F1	AUC
Random Forest	0.91	0.88	0.86	0.87	0.94
SVM	0.87	0.82	0.79	0.80	0.90
Logistic Regression	0.84	0.80	0.75	0.77	0.88
Neural Network (MLP)	0.89	0.85	0.82	0.83	0.92

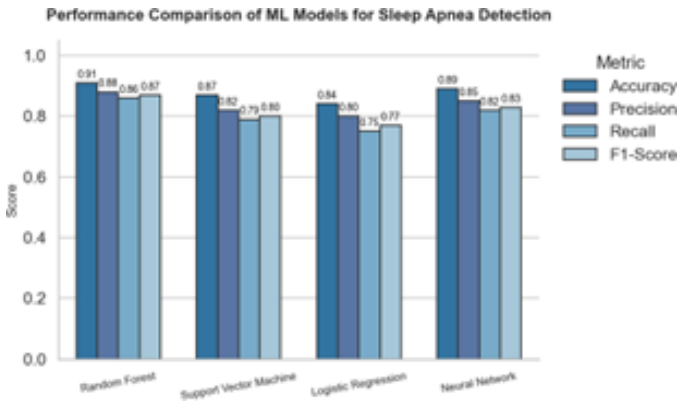


Fig. 3. Performance comparison of the evaluated machine learning models using Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Among the evaluated classifiers, the Random Forest model achieved the highest overall performance with an accuracy of 91%, F1-score of 0.87, and ROC-AUC of 0.94. These results indicate that the ensemble learning strategy effectively captures the complex nonlinear relationships present in ECG-derived features while maintaining robustness against noisy biomedical data and class imbalance. Although the MLP classifier also demonstrated competitive performance, it remained slightly inferior to Random Forest across all evaluation metrics. Logistic Regression produced the lowest performance because of its limited capability to model nonlinear feature relationships.

C. Confusion Matrix Analysis

The confusion matrix of the Random Forest classifier is presented in Table IV.

TABLE IV
Confusion Matrix of the Random Forest Classifier

Actual Class	**Predicted Apnea**	**Predicted Normal**
Apnea	130 (TP)	18 (FN)
Normal	25 (FP)	227 (TN)

The confusion matrix demonstrates that the proposed model correctly classified the majority of both apnea and normal ECG segments while maintaining a relatively small number of false-negative predictions. Since false negatives correspond to missed apnea events, minimizing this error is particularly important for clinical screening applications. The obtained results therefore indicate that the proposed model possesses high sensitivity and is suitable for preliminary sleep apnea detection.

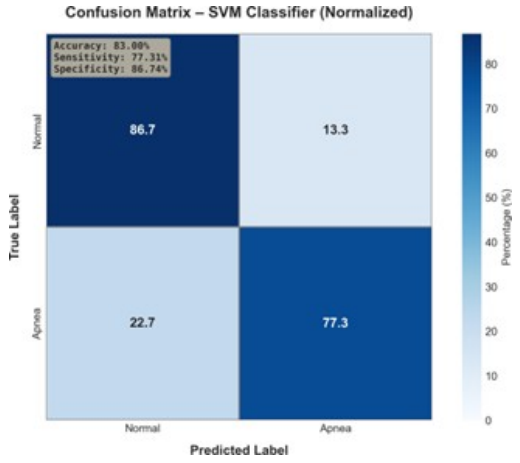


Fig. 4. Confusion matrix of the Random Forest classifier.

D. ROC Curve Analysis

Figure 5 presents the Receiver Operating Characteristic (ROC) curves obtained for all evaluated classifiers.

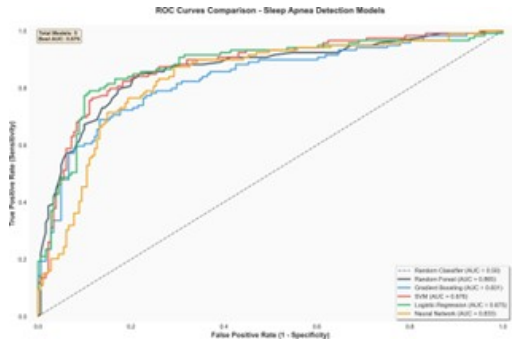


Fig. 5. ROC curves of the evaluated machine learning models.

The Random Forest classifier achieved the highest ROC-AUC value of 0.94, indicating excellent discrimination between apnea and non-apnea ECG segments across different decision thresholds. The MLP model achieved an AUC of 0.92, followed by SVM (0.90) and Logistic Regression (0.88). The superior ROC characteristics of the Random Forest model further confirm its robustness and generalization capability for sleep apnea classification.

E. Feature Importance Analysis

The feature importance scores generated by the Random Forest classifier provide insight into the physiological characteristics that contribute most significantly to apnea detection.



Fig. 6. Top ten ECG-derived features ranked according to Random Forest feature importance.

The five most influential features were:

- Signal Energy
- Standard Deviation
- Root Mean Square (RMS)
- Zero-Crossing Rate
- Heart Rate Variability (HRV) measures

These features capture variations in ECG morphology and cardiac rhythm that are commonly associated with apnea events, thereby providing meaningful physiological information for classification. Their high importance scores indicate that the proposed feature extraction strategy successfully captures discriminative characteristics required for reliable apnea detection.

F. Discussion

The experimental evaluation demonstrates that the proposed ECG-based machine learning framework provides reliable and computationally efficient sleep apnea detection. Compared with the other evaluated classifiers, Random Forest consistently achieved the highest performance across all evaluation metrics owing to its ensemble learning mechanism, robustness to noisy biomedical signals, and ability to model complex nonlinear relationships.

An additional advantage of the proposed framework is its interpretability. Unlike deep learning approaches that often operate as black-box models, the proposed feature-based methodology allows the contribution of each ECG feature to be quantified through feature importance analysis. This improves transparency and increases confidence in clinical decision support applications.

Furthermore, the trained model has been integrated into a Streamlit-based web application that enables clinicians to upload ECG recordings, perform automated apnea analysis, visualize ECG signals, and generate diagnostic reports in real time. This deployment demonstrates the practical applicability of the proposed framework for cost-effective and accessible sleep apnea screening.

VII. Conclusion and Future Work

In this paper, an AI-based Sleep Apnea Detection System has been presented, which analyzes electrocardiogram (ECG) signals and uses machine learning algorithms for non-invasive and automatic apnea detection. This system is able to distinguish between normal and apneic breathing patterns by extracting twelve time and statistical features from 30 second ECG segments. Among the classifiers tested – Random Forest, SVM, Logistic Regression, and MLP, the Random Forest classifier delivered the most accurate results in terms of accuracy, F1-score, and ROC-AUC [3], [9], [10].

Besides the high accuracy rate, the system stresses interpretability, efficiency, and clinical applicability. While deep learning models depend on extensive training data and computational power, this feature extraction technique provides a more efficient, transparent, and deployable method. Embedding the trained model into an interactive Streamlit web applica- tion increases its usability, enabling healthcare workers to up- load ECG signals, view the findings, and produce clinical re- ports instantly. Therefore, this system is a viable tool for sleep apnea pre-screening [14], [15].

Future Work: Although the system performs well, some areas can be enhanced in future research:

- **Inclusion of Advanced Features:** Advanced Feature Extraction: Applying frequency and time-frequency domain features like Fast Fourier Trans- form (FFT), Wavelet Transform, spectral entropy, etc. can detect more complex physiological patterns [2], [3], [6].
- **Deep Learning Integration:**Using CNNs or RNNs can facilitate end-to-end learning using raw ECG signals and may improve detection accuracy in large datasets [7], [8], [11].
- **Multimodal Data Fusion:** Combining ECG data with other physiological signals such as SpO2, respiratory effort, or airflow measurements can improve model robustness and reliability across various patient conditions [5], [9], [15].
- **Clinical Validation:** This system should be validated using real patient data in collaboration with hospitals for clinical reliability and compliance with medical standards [10], [13].
- **IoT and mHealth Integration:** Future iterations of this system can be expanded to mobile health (mHealth) and IoT wearable devices for continuous monitoring and apnea detection in non-clinical settings [14], [15].

In summary, the proposed Sleep Apnea Detection System demonstrates the potential of integrating machine learning with biomedical signal processing for healthcare applications. It provides a foundation for developing intelligent, affordable, and accessible tools for sleep disorder screening, contributing toward AI-driven preventive healthcare [1], [4], [10], [14].

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